

```

message");
    }
}

if(mq_close(mqd_sender) == (mqd_t)-1)
{
    perror("Sender: mq_close");
    exit(1);
}

printf("Sender: Exit\n");

exit(0);
}

// pmq_receiver_desc.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <mqueue.h>

#define MQ_QUEUE_PERMISSIONS 0660
#define MQ_MAX_MESSAGES 10
#define MQ_MAX_MSG_SIZE 128
#define MQ_MSG_BUFFER_SIZE MQ_MAX_MSG_SIZE + 10

int main (int argc, char **argv)
{
    mqd_t mqd_receiver, mqd_test;
    const char *postman_queue = "/mq-postman";
    const char *test_queue = "/mq-test";
    struct mq_attr attr;
    char rx_buffer[MQ_MSG_BUFFER_SIZE];

    attr.mq_flags = 0;
    attr.mq_maxmsg = MQ_MAX_MESSAGES;
    attr.mq_msgsize = MQ_MAX_MSG_SIZE;
    attr.mq_curmsgs = 0;

    if((mqd_test = mq_open(test_queue, O_RDONLY | O_CREAT,
MQ_QUEUE_PERMISSIONS, &attr)) == (mqd_t)-1)
    {
        perror ("Receiver: mq_open (test)");
        exit (1);
    }
    else
    {
        printf ("Test MQ %s opened with desc: %d\n",
test_queue, mqd_test);
    }

    if((mqd_receiver = mq_open(postman_queue, O_RDONLY | O_

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_CREAT, MQ_QUEUE_PERMISSIONS, &attr)) == (mqd_t)-1)
    {
        perror("Receiver: mq_open (receiver)");
        exit(1);
    }
    else
    {
        printf("Receiver MQ %s opened with desc:
%d\n", postman_queue, mqd_receiver);
    }

    int i=10;
    while(i)
    {
        printf("Receiver: blocked until message
arrives...\n");
        // get the oldest message with highest
priority

        if(mq_receive (mqd_receiver, rx_buffer, MQ_
MSG_BUFFER_SIZE, NULL) == (mqd_t)-1)
        {
            perror("Receiver: mq_receive");
            exit(1);
        }
        else
        {
            printf("Receiver: message received:
[%s]\n", rx_buffer);

            i--;
        }
    }

    if(mq_close(mqd_receiver) == (mqd_t)-1)
    {
        perror("Receiver: mq_close (receiver)");
        exit(1);
    }
    if(mq_close(mqd_test) == (mqd_t)-1)
    {
        perror("Receiver: mq_close (test)");
        exit(1);
    }
    printf("Receiver: Exit\n");

    exit(0);
}

```

You can compile the above two programs and any other program in this document by giving the following command in a terminal:

```
$gcc -lrt -o <executable_name> <c source file>
```